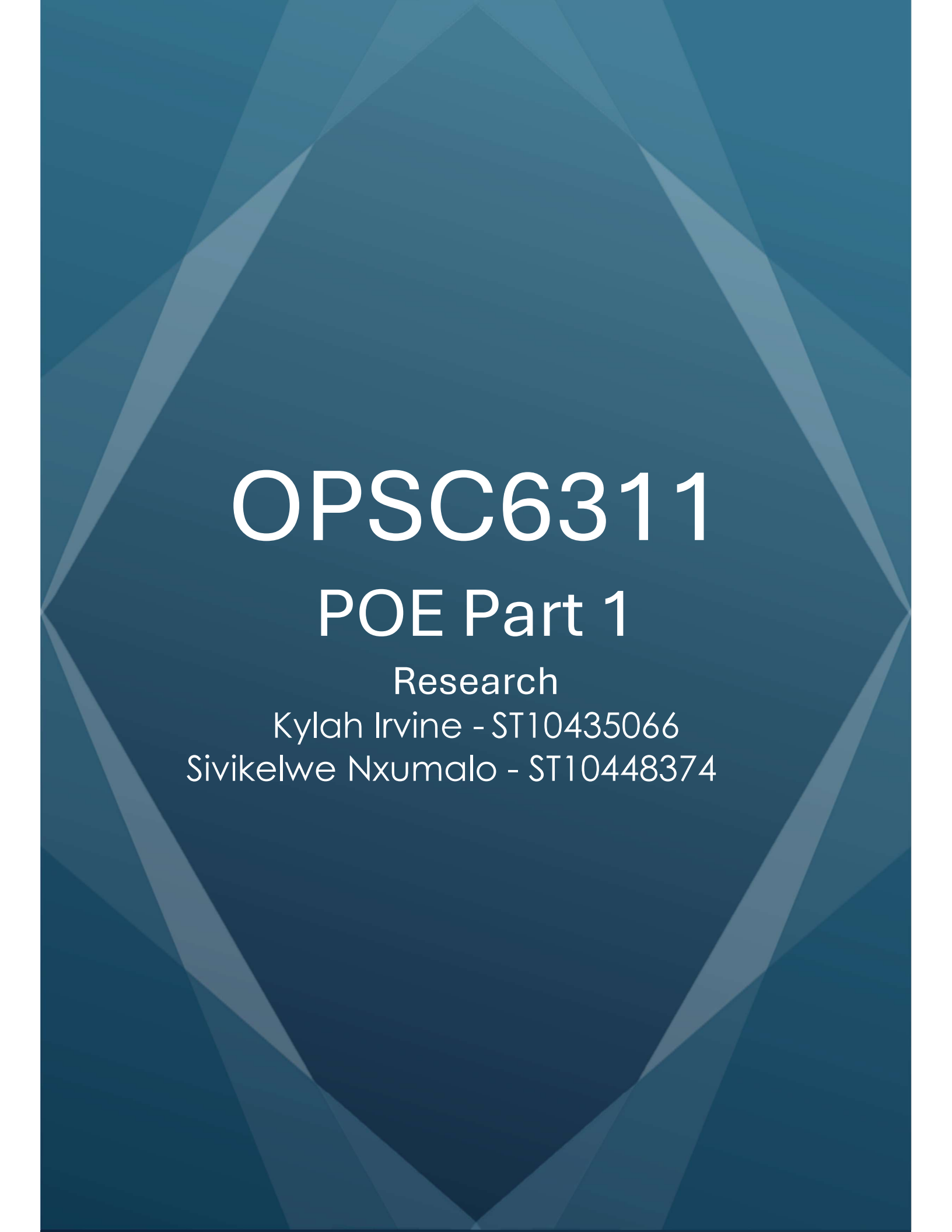




OPSC6311

POE Part 1

Research

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Introduction

Moving toward a cashless society has made it intimidatingly tough for people, particularly students, to keep track of where their money goes. Between quick tap-to-pay meals and endless subscriptions, it is easy to lose sight of a monthly budget before the first week is even over.

This report looks into three major players in the budget app market (Mint, YNAB and Buddy) to figure out precisely what kind of design and logic is currently needed to build a better budgeting tool for the Android platform. The goal this phase of the development process (part 1) is to take the best features of each application we have conducted research on and form an initial design. The end goal of the POE is to create an Android application, including the following features: the automation found in Mint, the strict discipline of YNAB and the social features of Buddy, of our own that specifically tailors to the South African student experience.

Analysis of Budgeting Apps

Mint

Overview

Traditionally the "gold standard" for free budgeting (Moneywise, 2023), Mint was known for its automated transaction syncing and powerful visual filters. According to multiple sources this app is rather comprehensive and highly praised (Moneywise, 2023; Policygenius, 2019). It is/was owned by Intuit, the same company behind TurboTax and QuickBooks (Investopedia, 2025). So the company has a decent reputation when it comes to financial programs.

The Android Gap

As of 2026, the original Mint experience is no longer available on Android. However, its legacy remains the benchmark for automated categorization and multi-filter spending views.

Strengths

- Exceptional at "set it and forget it" tracking
- High-quality data visualization (pie charts/bar graphs)
- Ability to link bank accounts to the app and have your financial data seamlessly integrated
- Notifications when you exceed your budget
- Automatically sorts transactions into categories
- Ability to see credit score

(Moneywise, 2023; Forbes Advisor, 2026; Policygenius, 2019, Investopedia, 2025)

Weaknesses

- Reactive rather than proactive. The transition away from a dedicated app has left a void for users who loved its specific filtering logic.
- Noted as overwhelming
- Notifications can become annoying
- Contains ads as they make their money via recommending you financial services
- Privacy concerns over Mint selling data
- Possible syncing issues

(Moneywise, 2023; Forbes Advisor, 2026; Policygenius, 2019, Investopedia, 2025)

Innovative Features

- Filtering transactions by date, category and tags simultaneously*
- Auto Categorizing
- Automatically pull data from banking app
- Ability to see credit score

*Features we plan to implement in our app

Interface analysis

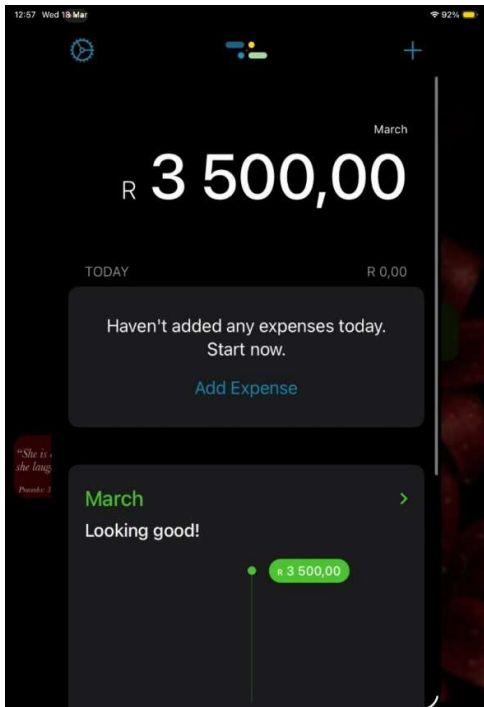


Figure 1 Mint app dashboard

As seen in Figure 1, Mint uses a "Dark Mode" aesthetic with a high-contrast balance display (R3 500,00). The "Looking Good!" status card is a vital psychological UX element that encourages positive financial behaviour.

The clickable elements of the UI are also easy to locate and somewhat intuitive as to what they do. The gear represents settings and the plus indicates an adding function. This aligns with how we perceive these symbols in a UI, with gears being used to indicate settings possibly originating from the idea that to fix or change a machine's function you "get all up in the gears and guts of the machine" for lack of professional phrasing. The plus sign is used most notably in maths to indicate that you are adding a new value to the existing one. Hence the symbol works for a UI to show you can add something to what you already have.

As seen in Figure 2, the app uses bar charts to compare current spending (R3 800,00) against historical averages (R1 266,67). This horizontal "average line" is a key feature we will adopt to help students see when they are overspending. It is also sorted by months so you can easily see which months tend to be the most expensive. This is a helpful trend to track as, for example, yearly subscriptions can often sneak up on us since we forget about the renewal date. If you are able to see the graph often it may make you think "why was that month so expensive?" and possibly you are reminded about those extra expenses. The categories graph is also very helpful as it will show what you are spending the most on. Especially if you create a category for something like "Entertainment" or "Treats" which are things we often overspend on without realizing.

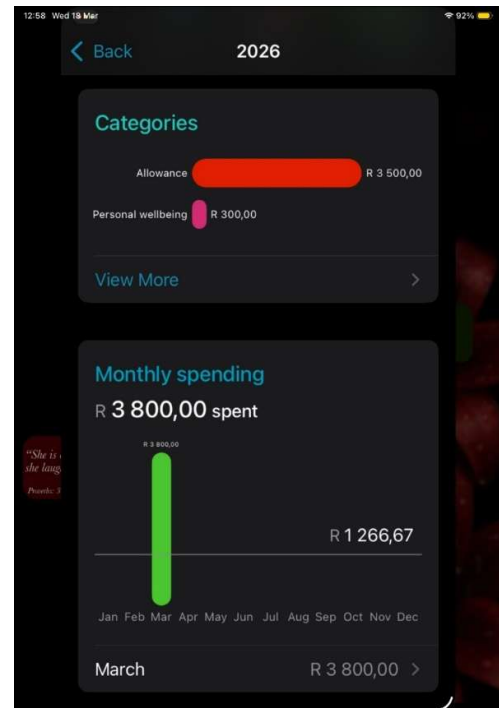


Figure 2 Mint app Insights

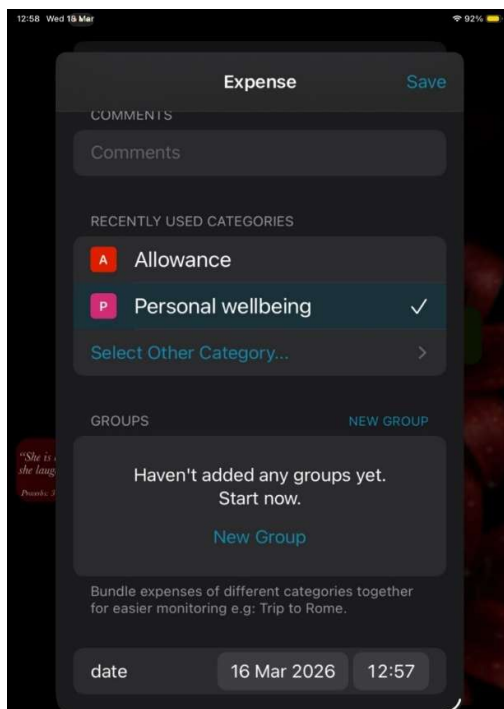


Figure 3 Mint app Add Expense

As seen in Figure 3, the "Add Expense" screen allows for "Groups." This is an innovative feature that lets users bundle transactions under a specific project (e.g., "Cape Town Weekend").

Being able to group categories together is an interesting feature. It would allow the user to, for example, group all income sources (Allowance, salary, etc) together to see how much money is coming in overall, while still allowing them to divide the sources.

You Need A Budget (YNAB)

Overview

A philosophy-driven app based on Zero-Based Budgeting, where every Rand is given a job/ category (NerdWallet, 2026; Mustachian Post, 2026).

The Cost Barrier

While YNAB is highly effective, its subscription model (R190–R250/month) is often too expensive for South African students (Mustachian Post, 2026).

Strengths

- Forces intentional spending
- "Age of Money" metric helps users break the pay-to-pay cycle (NerdWallet, 2026)
- Every Rand can be accounted for
- Encourages user to be proactive
- Users set goals for themselves
- Sends users reminders with actionable advice
- Linking to bank accounts (Mustachian Post, 2026; NerdWallet, 2026)

Weaknesses

- High learning curve (Mustachian Post, 2026)
- Manual entry can be tedious
- Expensive for those on a tight budget (especially students) (Mustachian Post, 2026)
- Cannot plan with money you don't have (future income/expenses)
- New users must do categorizing manually
- Not helpful with investments (Investopedia, 2026)

Innovative Features

- "Roll with the Punches" logic (balance overspending by reallocation)*
- Tracking/allocation every cent (no money does nothing)

*Features we plan to implement in our app

Interface Analysis

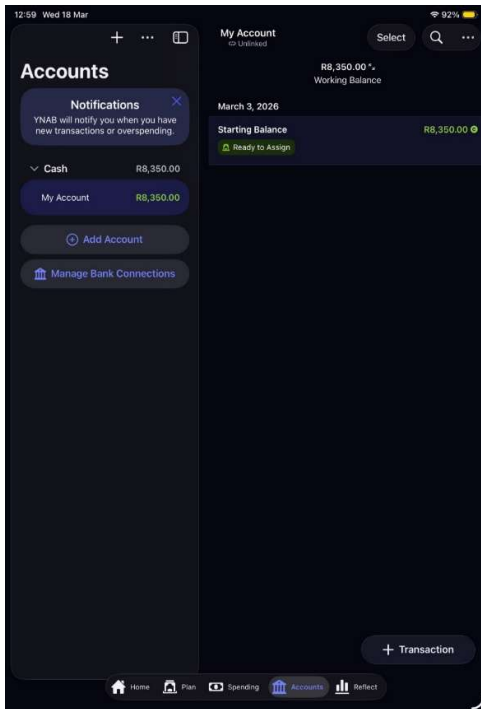


Figure 4 YNAB Initial Account Setup

As seen in Figure 4, this is the start of the budgeting cycle where the user links their accounts and adds their starting monthly income or current balance.

The user interacts by clicking "Add Account" or "Manage Bank Connections" to connect they're banking details to the app.

You can see the money amounts are displayed in green, making them easy to distinguish from the rest of the text.

As seen in Figure 5, once the income is visible, the user begins the process of "Giving Every Rand a Job" by selecting which specific bill the money should cover, adding the amounts from the total income to categories.

The user taps on a category like "Rent" or "Utilities" to open a selection modal, allowing them to funnel funds directly from their unassigned pool.

The background here is very visually interesting, with the blurry colour gradient. It makes it easy to see the borders of the boxes while still keeping with the dark mode theming.

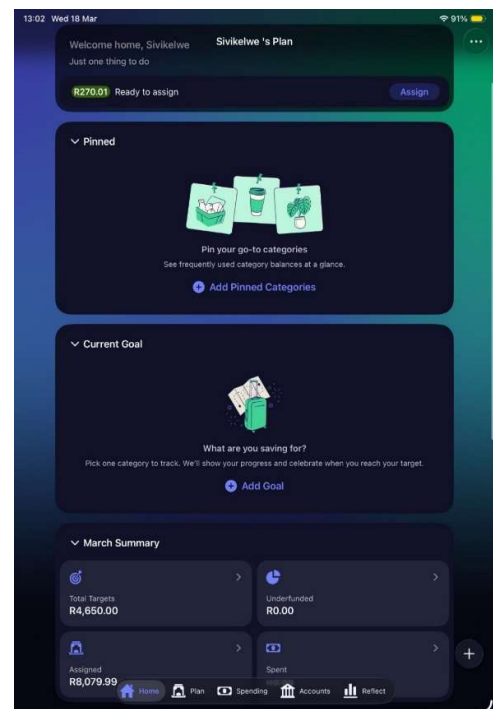


Figure 5 YNAB Income Allocation

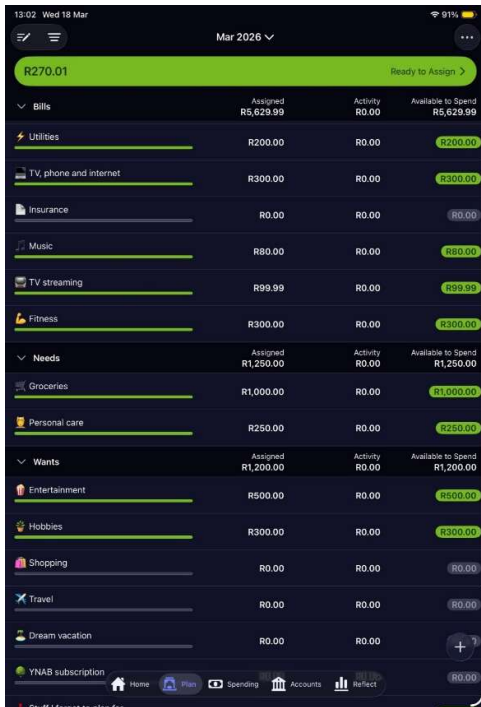


Figure 6 YNAB Budget Planning and Prioritisation

As seen in Figure 6, this view shows the user setting up their monthly spending plan and organising their income into "Needs" and "Wants" to see if they can cover the month.

The user scrolls through the category list and inputs specific amounts into each field.

As seen in Figure 7, the home screen acts as the user's daily check-in point to see what income still needs to be assigned before the month progresses.

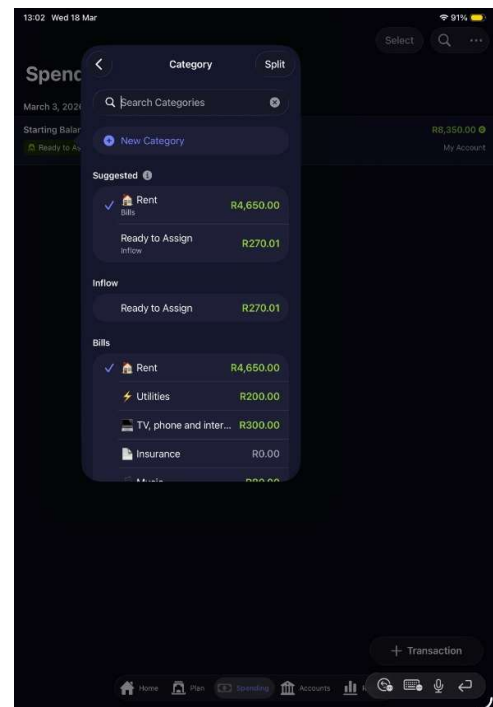


Figure 7 YNAB Dashboard and Task Management

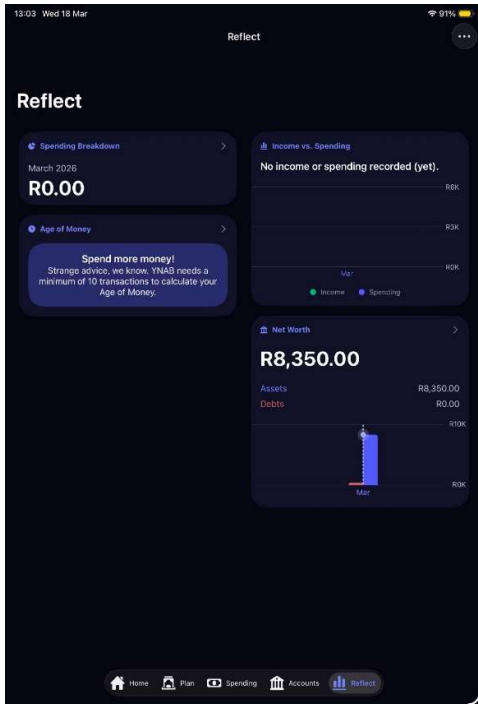


Figure 8 YNAB Reporting and Financial Reflection

As seen in Figure 8, the last step of the first setup where the user reviews their net worth and income-to-spending ratio to ensure their budget is sustainable. The graph is a good way to visualize how you're doing overall, and the idea of having a "reflect" screen may make users think about habits that aren't explicitly shown here.

Buddy

Overview

A highly aesthetic, community-favourite app known for its "Split" feature and simplicity (Financial IT, 2026).

Platform Constraint

Buddy has traditionally been an iOS favourite and is not natively available for Android Studio development, making it a perfect candidate for reimagining on the Android platform (Financial IT, 2026).

Strengths

- Excellent "Buddy" system for shared expenses (friends or roommates) (Financial IT, 2026)
- Offers a clean and non-intimidating UI
- Has themes to customize your experience
- Can link bank account
- Recurring transactions
- Solo-dev story appeals to Gen Z (Financial IT, 2026)
- Can upload Excel budgets
- Can use emojis (appeal to millennials)

Weaknesses

- Limited availability on Android (Financial IT, 2026)
- Some advanced features are locked behind a Premium paywall (Financial IT, 2026)

Innovative Features

- Shared Transaction is perfect for students in shared accommodation to track communal grocery or electricity costs.
- Themes*
- Ability to use emojis*

*Features we plan to implement

Interface Analysis



Figure 9 Buddy Household Overview

As seen in Figure 10, this screen helps the user map out their monthly income and expenses chronologically to ensure they have enough cash flow for specific dates.

The user taps on specific dates to add income (like the +R7,650 on the 15th) or track a high-spending day (like the 4.6k on the 1st), allowing for a time-based view of their budget.

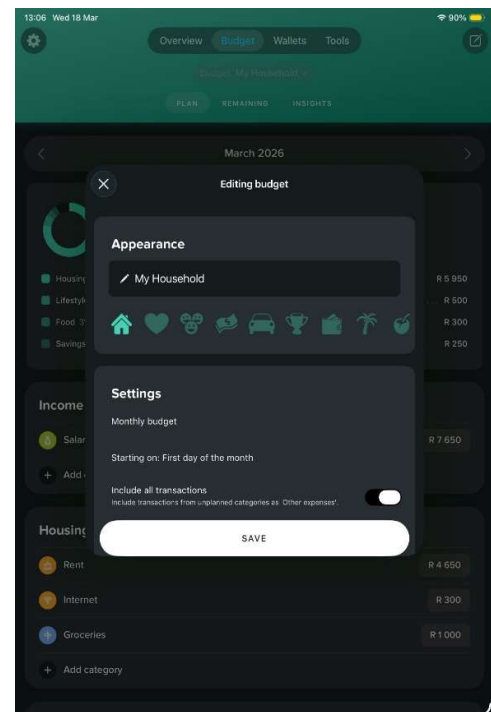


Figure 10 Buddy Spending Calendar

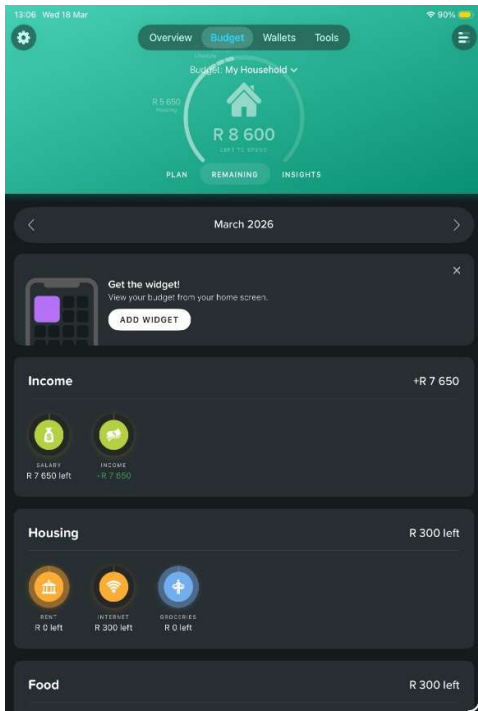


Figure 11 Buddy Transaction List

As seen in Figure 12, in this stage, the user creates their "Planned Expenses" by category (Housing, Lifestyle, Food) to see a percentage breakdown of where their salary is going.

The user taps "Add category" to build out their custom budget structure and types in the expected amounts for items like "Rent" or "Internet."

As seen in Figure 11, this is the logging stage where the user verifies that all starting transactions for the month (Rent, Salary, Groceries) are accurately recorded.

The user scrolls through the list to review specific entries and can tap the "+" button in the bottom corner to manually add any income or expenses that were not captured.

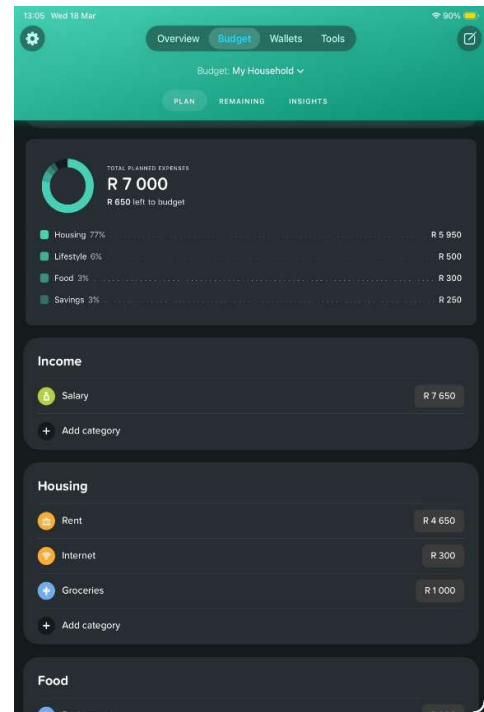


Figure 12 Buddy Budget Planning

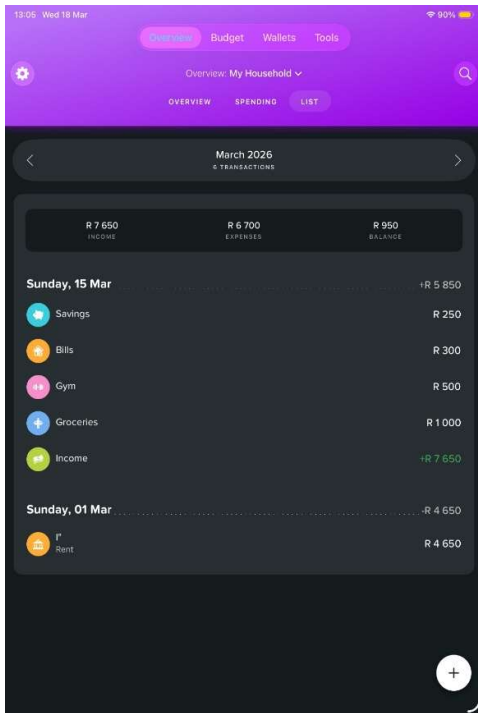


Figure 13 Buddy Remaining Budget

As seen in Figure 13, this view shows the "Left to Spend" amount, acting as a real-time guide for the user after income has been assigned.

The user looks at the circular icons to see which "envelopes" are empty (R0 left) and which still have funds available, helping them make quick spending decisions on the go.

As seen in Figure 14, this is the configuration step where the user names their budget and sets the start date for their financial month.

The user selects a personalised icon (like the house or heart) and toggles settings like "Include all transactions" before hitting "Save" to finish their budget framework.

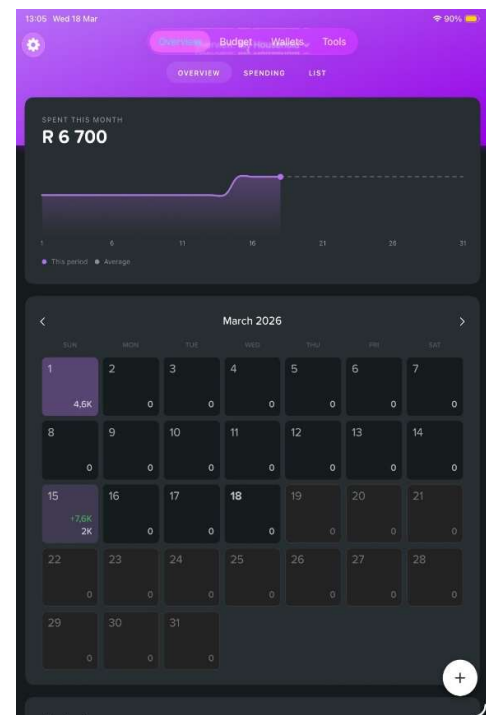


Figure 14 Buddy Budget Settings and Customisation

Selection Process & External Context

According to recent 2026 reviews from PCMag and FinanceBuzz, the shift in the app market has moved toward "Hyper-Personalization". Even though applications like Monarch Money and PocketGuard are rising stars, Mint, YNAB and Buddy were selected for this project specifically because they represent the three pillars of budgeting (automation (Mint), discipline (YNAB) and collaboration (Buddy)). By analysing these, our Kotlin application can extract the best of each while remaining free and accessible on Android.

Student Research Survey

To further validate the need for this app, a survey was conducted among 10 students at Student and living accommodation on 210 Loop Street Cape Town 8000.

Survey Questions:

1. Do you currently use a budgeting app? (Yes/No)
2. What is the number 1 reason you stopped using budgeting tools? (Too expensive / Too complicated / Forgot to log)
3. Would you prefer manual entry or bank syncing?
4. How important is a "shared expense" feature for roommates? (Scale 1-5)

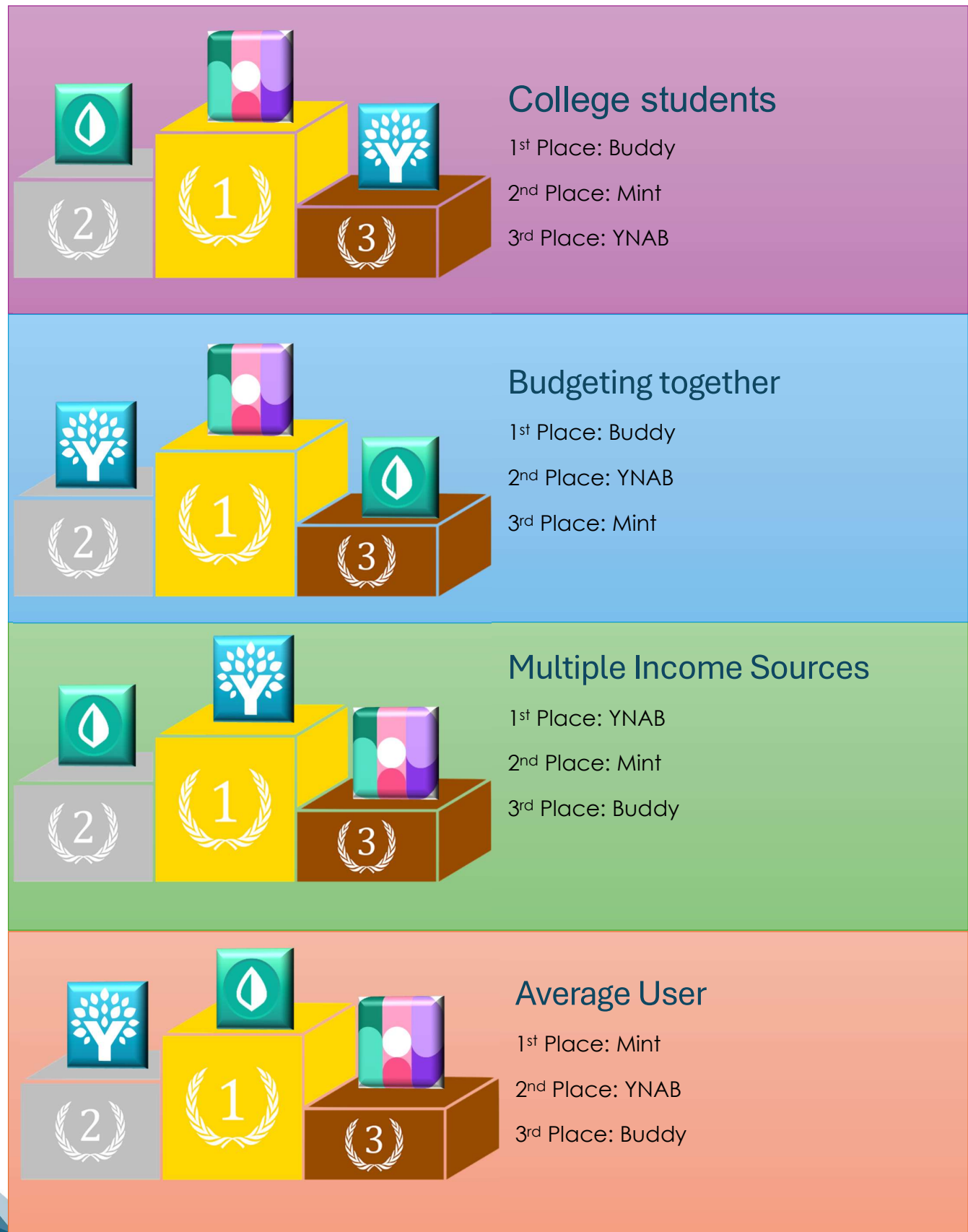
Results:

- 75% of students do not use a paid app like YNAB due to the cost.
- 60% cited "Complexity" as the main reason for quitting previous apps.
- Average rating of 3.5/5 for the importance of a shared expense feature, confirming that a "Buddy-style" feature is highly desired in the local student context.

This research indicates a clear market gap. A free, Android-native application that combines the advanced filtering of Mint, the goal-oriented discipline of YNAB, and the collaborative spirit of Buddy.

Infographic

Based on our research, we have ranked the 3 apps based on 4 types of users



Glossary (Appendix)

CRUD: Create, Read, Update, and Delete (the four actions the app performs on data).

HCI (Human-Computer Interaction): The science of making apps easy and comfortable for humans to use.

Neo-Minimalism: A modern design style that uses simplicity, soft shadows, and gradients.

Room Database: A secure "storage locker" on your phone for your financial data.

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